

Mcom - Sem IV
NEP 2020
11/7/25

Time: 2 Hours

Marks: 50

Q. 1. A) The deposit of Rs. 12,000 is made to earn interest @ 13% p.a. Find out the future value at the end of three year of this deposit if the compounding period is - (14)

1. Annually
2. Semi-Annually
3. Quarterly
4. Monthly

OR

Q. 1. B) Explain the consequences of poor cash management in personal financial planning. (07)

Q. 1. C) Write a note on money management with managing checking and savings account. (07)

Q. 2. A) Mr. Sharma, a 45 year old Mumbai professional, has seen his insurance costs spike due to inflation, a minor car repair that cost ₹5,000 in early 2023 now totals to ₹6,600 with rental and parts, while a similar medical emergency that once cost ₹8,000 in 2022 now bills ₹14,000. In India, medical inflation averages 14% annually, doubling treatment costs over five years and health insurance premiums have surged 50%-200% in just three years. Faced with broader pressures, from supply-chain disruptions driving up repair and auto prices to government proposals for a 10% hike in third-party motor insurance. Mr. Sharma now grapples with higher deductibles, increased premiums, and greater out-of-pocket exposure. This scenario highlights the pressing need for periodic policy reviews, higher deductibles or top-ups, and proactive purchase timing to help manage rapidly rising costs. (12)

- a. Analyse how inflation has affected Mr. Sharma's motor repair and medical expenses, and discuss the implications for his insurance coverage.
- b. Evaluate the impact of rising premiums, deductibles, and out-of-pocket exposure on Mr. Sharma's financial well-being and insurance decision-making.
- c. Propose and justify three risk-management strategies that Mr. Sharma could implement to manage health and motor insurance costs.
- d. Discuss the timing and frequency of policy reviews and insurer switching as a tool to optimize Mr. Sharma's insurance expenses in light of market trends.

OR

Q. 2. B) Rahul is 35 years old, earning ₹10 lakhs per annum. He wants to buy a term insurance plan. He expects his family to need ₹50,000 per month for the next 20 years in case of his absence.

- Assume an inflation rate of 5% annually.
 - Use a simple Present Value (PV) approach without compounding for simplicity.
- a) Estimate the total financial need of his family over 20 years in today's value.
 - b) Suggest an appropriate term insurance cover Rahul should consider based on this estimate.
 - c) Mention two additional riders that can enhance his coverage. (12)

Q. 3. A) Mr. Verma is 40 years old and wants to estimate the annual income he will need after retirement using the Replacement Ratio Method. (06)

- Current Annual Salary: ₹15,00,000
- Replacement Ratio: 80% (He wants to maintain 80% of his pre-retirement income)
- Expected Inflation Rate: 5% per annum
- Years until Retirement: 25 years

You are required to calculate:

1. The required annual income at retirement.
2. The inflation-adjusted amount he will need at the time of retirement

Q. 3. B) Write a note on Public Provident Fund. (06)

OR

Q. 3. C) Explain any three retirement need analyses technique. (06)

Q. 3. D) What are the various types of pension. (06)

Q. 4. A) Ravi, a 40-year-old investor, builds a ₹1 crore portfolio across 30 % domestic stocks, 20 % global stocks, 20 % bonds, 15 % REITs (real estate), 10 % mutual funds, and 5 % index-futures for modest hedging. He applies Modern Portfolio Theory (diversifying to reduce portfolio volatility) uses Hierarchical Risk Parity (HRP) to balance risk evenly across clusters of correlated assets rather than allocating by capital. Semi-annual rebalancing keeps allocations aligned. In back tests, this structure achieves stronger risk-adjusted returns, higher Sharpe ratios and smoother volatility, compared to a standard 60/40 equity-bond setup, with real estate via REITs offering added inflation protection.

(12)

- Evaluate how Ravi's portfolio aligns with the efficient frontier under Modern Portfolio Theory in terms of risk-adjusted return.
- Explain how Hierarchical Risk Parity's clustering and recursive allocation helps balance risk more effectively than mean-variance weighting.
- Analyze the role of index-future derivatives in providing downside protection and discuss associated risks like basis risk and cost.
- Discuss the pros and cons of semi-annual rebalancing, and propose an alternative rebalancing method (e.g., threshold-based) with rationale.

OR

Q. 4. B) Compare active and passive investment strategies in terms of cost, performance, flexibility, and tax implications, and discuss how strategic vs tactical asset allocation decisions influence portfolio outcomes. (12)
